

Composite Solids, Density and Design

Composite solid volumes, composite surface areas, density and mass, and design constraints

- Useful formulas: cylinder $V = \pi r^2 h$, curved surface $= 2\pi r h$; cone $V = \frac{1}{3}\pi r^2 h$, curved surface $= \pi r \ell$; sphere $V = \frac{4}{3}\pi r^3$, surface $= 4\pi r^2$; density = mass $\div$ volume.
 - After the first few warm-ups the methods are mixed — decide what the situation asks for before you reach for a formula.
 - Use the π key and round only at the final step, to the nearest hundredth.
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1. A grain silo is a plain cylinder of radius 4 m and height 7 m. Find its volume.

Answer: _____ m³

2. A concrete plinth is a rectangular block 1.2 m long, 0.8 m wide and 0.5 m tall, with a cylindrical hole of radius 0.2 m drilled straight through from top to bottom. Find the volume of concrete in the plinth.

Answer: _____ m³

3. An open-topped cylindrical water tank has radius 2 m and height 6 m. Find the area of sheet metal needed for its curved wall and its base.

Answer: _____ m²

4. A block of oak has volume 2400 cm^3 and density 0.75 g/cm^3 . Find its mass.

Answer: _____ g

5. A farm silo is a cylinder of radius 3 m and height 8 m, topped by a hemisphere of the same radius. Find the total volume of the silo.

Answer: _____ m^3

6. A bronze statuette has mass 5.4 kg, which is 5400 g, and bronze has density 8.8 g/cm^3 . Find the volume of the statuette.

Answer: _____ cm^3

7. A cylindrical water tank must hold exactly 15 m^3 . The site allows a radius of only 1.5 m. Find the height the tank must be built to.

Answer: _____ m

8. A grain hopper is a cone of radius 5 cm and slant height 13 cm sitting on a cylinder of the same radius and height 9 cm. The cylinder's bottom is closed; the cone and cylinder meet with no exposed disc between them. Find the total outside surface area of the hopper.

Answer: _____ cm^2

9. A glass paperweight is a cube of edge 6 cm with a hemispherical hollow of radius 2 cm scooped out of one face. The glass has density 2.5 g/cm^3 . Find the mass of the paperweight.

Answer: _____ g

10. A cylindrical grain bin of radius 2.5 m must hold 60 m^3 . It is built from steel weighing 18 kg for each square metre, and only the curved wall and the flat base use steel.

(a) Find the height the bin must be built to.

Answer: _____ m

(b) Find the mass of steel the bin needs.

Answer: _____ kg

(c) The capacity must stay at 60 m^3 . Explain why doubling the radius would not halve the mass of steel needed.